

Saurabh Kumar Singh

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Right to work in the UK: No employer sponsorship required

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Professional Summary

Instrumentation engineer and experimental physicist with strong experience in laser diagnostics, Raman spectroscopy, optical metrology, and precision measurement system development. Experienced in designing, building, aligning, and calibrating custom spectroscopy and diagnostic instruments for real-time monitoring and high-accuracy experimental applications. Hands-on expertise in optical system integration, interferometry, detector instrumentation, vacuum-based experimental systems, and automation using Python, LabVIEW, and MATLAB.

Core Competencies

- **Laser and Optical Diagnostics:** Raman spectroscopy, laser frequency stabilisation, interferometry, optical alignment, beam diagnostics, precision wavelength measurement
- **Instrumentation Development:** End-to-end design and construction of custom diagnostic instruments, spectrometer development, opto-mechanical integration, detector systems, fast electronics
- **Spectroscopy and Measurement Systems:** Real-time optical monitoring, mass spectrometry, TOF-MS, quadrupole mass spectrometry, velocity map imaging
- **Simulation and Analysis:** SIMION, MolFlow+, MATLAB, Python, signal processing, experimental calibration, image reconstruction
- **Automation and Control:** LabVIEW, Python-based instrument control, Arduino/ESP32 microcontroller programming, data acquisition and hardware interfacing
- **Vacuum and Hardware Engineering:** UHV design, conductance calculations, flange and piping layouts, leak checking, system integration

Relevant Instrumentation Projects

Raman Spectrometer Prototype for Real-Time CVD Monitoring and SERS Applications

- Contributed to the design and development of a prototype Raman spectrometer for real-time monitoring of CVD growth in an industrially relevant measurement context.
- Developed the spectrometer platform with dual purpose: in-situ process diagnostics for CVD growth monitoring and experimental use in surface-enhanced Raman spectroscopy (SERS) studies.
- Performed prototype calibration using known neon emission lines and a tunable dye laser source.
- Supported optical system development, alignment, and spectroscopic validation for real-time measurement applications.

Precision Wavemeter Based on Crossed-Fringe Fizeau Interferometer

- Designed and built a high-precision optical wavemeter for laser characterisation and frequency monitoring.
- Implemented thermal drift compensation and precision optical alignment strategies.
- Achieved part-per-billion level accuracy and contributed to patent filing and publication.

TOF Mass Spectrometer and Ion Optics Development

- Designed and built a complete time-of-flight mass spectrometer based on McLaren space-focusing geometry.
- Performed ion optics optimisation using SIMION for extraction region, lens voltages, and flight geometry.
- Experimentally validated instrument performance against simulation predictions.

Velocity Map Imaging and Detection Systems

- Developed ion imaging and velocity map imaging assemblies for state-resolved scattering experiments.
- Calibrated MCP/phosphor/CCD-based detection systems and image reconstruction workflows.

Quadrupole Mass Spectrometer Integration

- Installed, configured, and calibrated Hiden and Pfeiffer quadrupole mass spectrometers.
- Performed sensitivity calibration, background correction, dwell-time optimisation, and signal-to-noise improvement.

Professional Experience

Postdoctoral Research Assistant

Oct 2024 – Present

Surface Dynamics Lab, Swansea University, Wales, UK

- Designing, installing, and calibrating mass-spectrometry-based diagnostic systems for helium atom scattering and molecular beam experiments.
- Developing a compact high-flux helium source for molecular interferometry using CAD design, conductance modelling, and MolFlow+ simulation.
- Integrating quadrupole mass spectrometers into experimental platforms for real-time detection and surface reactivity studies.
- Supporting development of high-precision experimental hardware for gas-surface interaction studies under ultra-high vacuum conditions.

Ph.D. Researcher

2018 – 2024

Tata Institute of Fundamental Research (TIFR), Hyderabad, India

- Designed and built multiple custom optical and ion-detection instruments for spectroscopy, molecular beam, and surface science experiments.
- Developed a high-precision wavemeter based on crossed-fringe Fizeau interferometry for laser measurement and characterisation.
- Built and optimised TOF-MS and velocity map imaging systems using SIMION-based ion optics simulations and experimental validation.
- Contributed to the design and calibration of a Raman spectrometer prototype for real-time CVD growth monitoring and SERS research applications.
- Worked on laser-based diagnostics, optical alignment, detector integration, and automated data acquisition workflows.

Technical Tools

Programming / Control	Python, LabVIEW, MATLAB, Arduino, ESP32
Simulation / Modelling	SIMION, MolFlow+, CAD tools, conductance modelling
Instrumentation	Raman spectroscopy, interferometry, TOF-MS, QMS, VMI, MCP detectors, CCD-based imaging
Experimental Systems	UHV systems, molecular beams, laser diagnostics, detector electronics, vacuum hardware integration

Education

Ph.D. in Chemical Science	2018 – 2024
Tata Institute of Fundamental Research (TIFR), Hyderabad, India	
Thesis: <i>Understanding the Dissociation Dynamics of CO₂ on Cu(110) and Developing Techniques for State-Selected Molecule-Surface Scattering Experiments</i>	
M.Sc. in Physical Chemistry	2016 – 2018
Banaras Hindu University, India	
B.Sc. (Hons.) Chemistry	2012 – 2015
Banaras Hindu University, India	

Patent

Saurabh Kumar Singh, Avinash Kumar, and Pranav R. Shirhatti
A multistage thermal drift compensated cross-fringed Fizeau interferometer wavemeter and method thereof.
Indian Patent Application No. 202321048766, filed 20-Jul-2023.

Selected Publications

- Saurabh Kumar Singh, Avinash Kumar, and Pranav R. Shirhatti, *A simple and low-cost setup for part per billion level frequency stabilization and characterization of red He-Ne laser*, HardwareX, 2023.
- Geetika Bhardwaj, Saurabh Kumar Singh, and Pranav R. Shirhatti, *A compact and highly collimated atomic/molecular beam source*, Review of Scientific Instruments, 2023.
- Full publication list available via ORCID: [0000-0002-2760-9671](https://orcid.org/0000-0002-2760-9671))

References

Available upon request.